Amitesh Puri

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Education

Generative AI – Principles and Applications — IISc Bangalore
Grade: A+ — May – Jul 2026

Mathematical Foundations of Machine Learning — NPTEL · IISc Bangalore
Score: 68% — Jan – Apr 2026

Manipal Academy of Higher Education — Udupi, Karnataka, India
M.Sc. in Applied Mathematics and Computing *CGPA: 7.28 / 10* — Aug 2023 – Jun 2025

Ramnagar College, Vidyasagar University — Paschim Medinipur, West Bengal, India
B.Sc. in Mathematics *Final Grade: 6.21/10* — Jul 2017 – Oct 2020

Publications

Message Passing Graph Neural Networks: A Study — 2025
*Simran, **A. Puri** and S. Akansha · Global and Stochastic Analysis, 12(1), 80*

Research Experience

Physics-residual Flow Matching under distribution shift [Report] [GitHub]
Independent research report

A common argument for physics-constrained generative models is that respecting the governing equation should keep them reliable when the data moves away from training. This experimental study examined this on a next-step operator for 2D Navier-Stokes, comparing plain flow matching with a weighted physics residual and a conflict-free gradient variant across increasing shift severity and sampling budget. In distribution, the weighted residual lowered relative error by 37 % and the PDE residual by 44 % over five seeds, and matched the baseline's 64-step accuracy in 4 to 8 steps, while the conflict-free variant could not be separated from the baseline on any in-distribution metric. The advantage did not survive higher-wavenumber initial conditions or stronger forcing. Under both shifts the residual model degraded about twice as much as the baseline relative to its own in-distribution error. At the strongest initial-condition shift it was 33 % less accurate than the baseline, and under the strongest forcing its rollouts failed at step 9.6 against 15.6. Even with the full governing equation built into the residual, the gain held only in distribution and within a narrow range of residual weights. The inductive bias of the residual is only as useful as the physics it encodes, which makes it unlikely to help where the equation is unknown or only partly specified, as in coupled multiphysics systems.

Master's Research

Addressing Conditional Shift in Graph Data: An Empirical Study of Distance Metrics [Report] [GitHub]
Master Research Report

GNNs trained on a labelled subset that does not represent the whole graph lose accuracy on the remaining nodes, a conditional shift between training and full distributions. The project added KLD, JSD, MMD and CMD to the CE loss as regularisers, alone and in weighted pairs, and used conformal prediction to measure reliability through prediction sets with a coverage guarantee. Over five architectures and 15 runs on Cora, every CMD-based regulariser raised accuracy under bias for all architectures except GAT, by up to 4.3 points. The largest recovery was GraphSAGE with CMD, from 0.692 to 0.736 against 0.808 without bias, which won back 37% of the accuracy lost to the bias, the most of any model on either dataset. The same regularisers cut the average prediction set by 30 to 45% for all except GCN at the 90% coverage target, while KLD, JSD and MMD alone stayed near the CE baseline. CMD alone performed close to the paired versions on most architectures, so it carried most of the gain, and the effect was weaker on Citeseer. Since these measures only align the distributions of training and test representations, with no model of what produces the shift, the results suggest that alignment agnostic to the source of the shift can narrow the generalisation gap but not close it.

Technical Training

Guided Projects Student — Nov 2022 – Feb 2023
The Machine Learning Company — Remote

Completed a set of applied projects across NLP and computer vision, building end-to-end pipelines for text classification, sentiment analysis, and language generation, and training and evaluating deep learning models for image classification and object detection.

Data Science Trainee — Aug 2021 – Jan 2022
Evarcity — Bangalore, India

Built Python-based data analysis and machine learning workflows on real-world datasets, with hands-on work in feature engineering, model evaluation, and data visualisation.

Technical Skills

Programming	Python, R
ML / DL Frameworks	PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, PyCaret
Data Analysis	Pandas, NumPy, Matplotlib, Seaborn, Data Visualization
Tools	Git, Gradio, Streamlit, Hugging Face Hub, Argilla